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Research Article

Keywords: Agriculture, IoT, machine learning, artificial intelligence, cloud service, data storage, irrigation, and farming

Posted Date: March 18th, 2024

DOI: https://doi.org/10.21203/rs.3.rs-4089574/v1

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Additional Declarations: No competing interests reported.

INTELLIGENT IRRIGATION SYSTEM WITH SMART FARMING USING ML AND ARTIFICIAL INTELLIGENCE TECHNIQUES

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Abstract

Due to increase in population, there is a drastic changes in whether condition which also affects agricultural productivity which leads to raise in food demand and creates a significant problem for humanity. This study suggests an automated control system of smart farming uses artificial intelligence, IoT and machine learning to monitor and regulate several agricultural regions vital to the entire farming process to solve lower productivity. Finally, provide better visualization and monitoring of the environment around us, we offered a web application that combines the different data produced by the sensors with the forecast from our models. The ensemble learning algorithm (SVM-KNN) to make an accuracy rate of 90% rather than individual SVM and KNN was 72% and 76%, respectively.

Keywords: Agriculture, IoT, machine learning, artificial intelligence, cloud service, data storage, irrigation, and farming.

1. Introduction

Agriculture is very crucial to human beings and the development of our country. The production of crops rises in direct proportion to the daily increase in the world's population. Agriculture production is primarily seasonal. More individuals are turning to this area because of smart agriculture, which we have recently heard more about. Contrary to popular belief, plants require the vitamins and minerals that soil offers for growth, not dirt. Plants also require light, water, oxygen, and carbon dioxide to survive and develop. In smart agriculture, it is possible to grow fruits, vegetables, and flowers comfortably using nutrient solutions that already contain the minerals the plants require. By doing this, plants won't have to search for the minerals they need in the entire soil, which will result in much more effective and fresher production. Agriculture will continue to take up the bulk of the world's freshwater resources85% of all resources due to population increase and growing food consumption. Plans based on science and technology must be developed immediately. The success of theoretical research contributions over the preceding ten years has made the desire for IoT in farm applications a reality. Agricultural sectors have recently implemented information technology (IT) solutions, such as technical, agronomic, managerial, etc., as part of precision agriculture techniques that help efficiently use water. In order to decrease water usage in a range of crops, from low-tech to high-tech, irrigation systems have been the focus of several research projects. The Internet of Things (IoT) provides a platform for "smart agriculture," wirelessly connecting a variety of sensors (including temperature, humidity, and soil moisture), a variety of hardware

(including water pumps), and data-analytical programs to assist farmers in better addressing complex agricultural problems like yield prediction and other issues throughout the entire growing and harvesting cycle.

Physical/mechanistic methods have lost ground to data-driven machine learning (ML) models for irrigation as the go-to technique for estimating soil moisture content in recent years. This is because the calibration of data-driven models is less data-intensive than the calibration of mechanistic models. Furthermore, ML models handle geographical and temporal data rapidly and easily, and they beat physical models in terms of prediction. By comparing pre-processed data to measured real-time data and utilizing the system's own feedback as input, the closed-loop controller employed by ML model-based irrigation systems learns from calibration.

The farmers' conventional techniques are now insufficient to meet the demand. In order to achieve these goals and provide many individuals in this region with fantastic work opportunities, a variety of inventive automation solutions have been devised. Artificial intelligence has become one of the most important technical developments across many sectors, including agriculture, banking, education, and robots. It is transforming the agricultural business and having a significant impact on it. AI protects the agriculture industry from a range of obstacles, including concerns about food safety, population expansion, climate change, and labour shortages. AI has allowed modern agriculture to progress to a new level. Artificial intelligence has enhanced crop production, real-time monitoring, harvesting, processing, and marketing. A number of high-tech computer-based methodologies have been developed in order to discover a range of important characteristics, including weed identification, yield detection, crop quality, and many more.

Objectives of proposed work

- To create an ensemble learning algorithm, which is used in irrigation systems with the help of current sensor data from agriculture.
- To create IoT based application with the help of NodeRED programming language and mosquito broker, which is used to monitor the sensor data and predicted results of rainfall and crop yields.
- To detect the rainfall prediction using the LSTM algorithm to predict the rain values in a particular time period from a webpage.
- To evaluate crop yield prediction using the dataset updated in the flask app with the help of a learning algorithm.

2. Related work

Smart farming and smart irrigations are both important in agriculture to help a farmer to increase the quality and quantity. Smart farming is developed using modern technology such as IoT (Internet of Things), ML (Machine Learning) and AI (artificial intelligence). Many factors affected crop growth. This chapter discusses how to develop farming for different applications.

2.1 Wireless sensor network and IoT applications for smart farming

Roopa G K, and Radhika Shetty, (2019) Comparing precision agriculture to traditional approaches, improved productivity and better resource management are expected to result in reduced costs and higher yields. Climate change is making water resources limited. Thus, it is critical to irrigate more effectively to maximize water utilization. With the development of the Internet of Things and the broad adoption of the Web in agriculture, wireless sensor networks are becoming more and more crucial. Making use of web services for oversight will be wise. Data is sent to the server by the microcontroller, which then gets instructions to start the irrigation system. Additionally, it does direct sensor data processing to conduct automated watering. Two pumps are used in the irrigation pumping system, and they are both relay-activated. For monitoring and setting up an irrigation schedule, a web application is employed. Additionally, it is utilized to specify WIU threshold values for automatic scheduling. Storage is performed using a SQL database.

Mohammad Wasi Rasooli, et al, (2020) IoT devices not only gave farmers new skills, but they also improved their time management. Sensors are particularly important for connectivity and data collecting. Farmers now spend less time in the field because IoT-powered tools are doing the work for them. It lessened the amount of work that farmers had to do and the number of resources that were wasted, including water used for irrigation and energy used to power so many electrical gadgets. The real-time applications may be readily monitored with IoT devices. IoT has the power to reshape the planet and enable intelligent agriculture.

Pankaj Kumar Kashyap et al, (2021) Precision irrigation may provide more, better, and more lucrative crops with fewer resources. To use water more effectively, many irrigation models based on machine learning have been developed. These models' poor capacity for learning makes them unsuitable for unstable climates. In this regard, this study recommends an intelligent irrigation system for precision agriculture that is backed by the IoT and built on deep learning neural networks (DLiSA). The mean and standard deviation are calculated exclusively for normalization reasons using data from the training period, and daily data are averaged, the standard deviation is subtracted, and the result is divided by the standard deviation. In order for the suggested LSTM -RNN model to learn effectively, this normalizing step is required.

Ajayi O et al. 2022, Measurements, such as the Water Quality Index (WQI) and Irrigation WQI (IWQI), are used to express the quantity of these qualities and ascertain the total water quality. This research proposes a network architecture to automatically determine if water samples are appropriate for drinking and irrigation by gathering real-time data on water attributes.

Alghazzawi. D et al. 2021, The greenhouse simplifies planting, which benefits agriculture in a number of ways. In agricultural modelling, gas sensors and soil pH sensors are widely used. These sensors might be applied in a number of agricultural IoT applications. The method covers the hardware design and functioning of the proposed model. There are also descriptions of several other agricultural evapotranspiration models. The Penman-Monteith equation investigates crucial topics, including congestion management. The irrigation system maintains the moisture level in soil using the sensory and control system during environmental conditions explained by **Amin A. B et al. 2021**

Badrun et al. 2021 The irrigation industry may benefit from the Internet of Things (IoT) innovative solutions, which can enhance irrigation management and lower operating infrastructure maintenance costs. One of the most important problems with intelligent irrigation systems is the inability of stakeholders to convert available data into comprehensive, trustworthy information that can be utilized in decision-making.

P. Pandiaraj et al, (2021) Agriculture uses machine learning algorithms to raise the quality and output of agricultural products. Using image recognition technology, pest control professionals can recognize and deal with various pests and rodents. One machine learning (ML) approach, the random forest algorithm, is used to identify pests and diagnose issues in four crops such as maize, potato, tomato, and wheat.

Dusan Markovic et al, (2021) Some pests, like Helicoverpa armigera, may change their appearance depending on the weather, temperature, and relative humidity. This work suggests an ML model that can estimate the daily presence of insects through a season while taking air temperature and relative humidity into consideration. A number of machine learning classification methods were used, and their accuracy in predicting the presence of insects was demonstrated (up to 76.5%).

2.2 AI-based smart farming techniques

Maria Teresa Linaza et al., (2021) By monitoring conditions and increasing production, artificial intelligence (AI) technologies help farmers make better decisions at the farm level. They are able to enhance yields while consuming less water and emitting fewer greenhouse gases because they are able to put on the proper number of inputs for each crop. Future applications of this study will incorporate fresh ideas for efficient livestock management as well as the recovery of plant and soil samples utilizing intelligent, autonomous robots.

Noon et al., (2021) Deep learning (DL) algorithms have been used in similar thorough studies on how to identify plant leaf stress. Based on the kind of stress, the size of the dataset, the training/test size, and the deep network employed, the approaches were separated into those for vegetables, fruit, and other crops. The identification and classification of all stress levels to find infectious diseases and the application of algorithms to various crops are two areas that need additional research. AI and robots are crucial in assisting or replacing manual involvement. On the one hand, robotics has been crucial to agricultural production and administration, and nowadays, robots carry out a variety of tasks independently, including weeding, watering, and monitoring specific plants. In-depth summaries of current robotic

grippers, gripping methods, strategies for sensors, and their applications in robotic agricultural tasks have been provided by **Zhang et al (2020).**

Akanksha et al., (2021), Four steps make up the methodology: preprocessing, feature extraction, segmentation and classification. The noise-containing photos are first eliminated before the images are transformed to RGB format. The feature extraction step is then assigned the R band. In order to categorize a picture as normal or abnormal, the classifier is fed the relevant properties. An Optimized Probabilistic Neural Network (OPNN) is used for classification. Artificial jelly optimization (AJO) is an approach that enhances PNN classifier performance.

Anand et al., (2021) Imbalance the agriculture, including standing water and weed patches, interfere with farming operations, resulting in incorrect utilization of the farmland and upsetting agricultural planning. Modern agricultural techniques might be more effective by monitoring fields and crops using Internet of Things (IoT)-enabled intelligent technologies. Remote sensing using Unmanned Aerial Vehicles (UAVs) is an effective method for gathering detailed photographs of farms. There is a lot of declaration for improving agricultural outputs by recognizing crop trends through the development of AI-assisted farming models employing visual data analytics for automatic pattern recognition from the gathered data.

Bhat et al. 2021 Precision agriculture use big data (machine learning, deep learning, etc.) because of its ability to analyse vast amounts of data and crucial abstract information and help agricultural practitioners understand farming operations and make accurate judgments. The primary goal of this remark is to inform readers about the most recent Big Data applications in smart agriculture and the corresponding social and financial concern.

Alharbi. H et al. 2021, the IoT, cloud, ambient, edge, artificial intelligence, and agricultural robotic systems are used for the real-time tracking, analysis, and processing of many processes and services. Additionally, these technologies can increase farm output and exceed future expectations by making agricultural processes more intelligent and resourceful through the use of automated systems and eliminating human involvement. And also introduced a hydraulic system to control the irrigation depended on Takagi-Sugeno fuzzy (TSF) model to reduce water conservation **Chrouta et al., (2019)**

Cioffi et al. (2020) Initialized innovation and adaption are vital in the industrial sector. This research was meant to pave the way for the use of cutting-edge technologies in agricultural production. To achieve sustainable growth, manufacturing needs worldwide prospects for agriculture smart production technology. As a result, the suggested work thoroughly examined research reports on machine learning (ML) and artificial intelligence (AI) in the sector.

Melović et al., (2020) This research was investigated to determine the critical elements that influence how effectively green customers purchase items from organic agriculture. Health benefits were recognized as the main buying trend from a certain production process by the relative importance of the index.

Shamshiri et al., (2020), to deal with production unpredictability and boost earnings, a microclimate study was done on the commercial items used in closed-field agriculture. In this

study, an Internet of Things application was proposed to assess model-based microclimate parameter values in agricultural greenhouse systems. Thanks to the IoT sensor node and simulation model, farmers now have a better understanding of the processing that occurs in crop-growing environments.

Shaikh et al., (2021) Artificial intelligence (AI)-enabled smart agriculture is assisting in ensuring the long-term profitability of agriculturalists. AI is utilized in a variety of fields, including managing soil and irrigation systems, predicting the weather, monitoring plant development, predicting diseases, and managing animals.

Kumar et al., (2021) To ensure the long-term survival of the sector, smart agriculture must utilize artificial intelligence (AI). These include controlling irrigation and soil systems, making weather forecasts, raising plants, identifying illnesses, and controlling animals. Our research has a strong emphasis on AI algorithms, especially how they affect system performance. We research novel AI techniques of different sizes.

2.3 Rainfall prediction with datasets

Nevavuori,et al, (2019) On environmental datasets from many countries, a lot of research has been done to predict rainfall using data mining and machine learning techniques. The nation's unequal rainfall distribution has an effect on agriculture, which is the cornerstone of the economy. In order to reduce the problem of the country's drought and flooding, it is crucial to plan for and implement the efficient use of rainfall water. The regional meteorological office in Bahir Dar City, Ethiopia provided the dataset, which was used to compare the performance of three machine learning techniques (Multivariate Linear Regression, Random Forest, and Extreme Gradient Boost). The Mean Absolute Error and Root Mean, Squared Error approaches were used to assess the performance of the machine learning model.

Nana Kofi Ahoi Appiah-Badu et al, (2022). Due to recent climate change and fluctuation, it isn't easy to estimate rainfall accurately. The effectiveness of classification systems for predicting rainfall has soared. The study supports the application of several categorization algorithms for rainfall forecasting throughout Ghana's diverse natural zones. Examples of classification methods include Decision Tree (DT), Multilayer Perceptron (MLP), Random Forest (RF), Extreme Gradient Boosting (XGB), and K-Nearest Neighbor (KNN). The data was given by the Ghana Meteorological Agency and covered the period from 1980 to 2019. It has a range of climatic characteristics. The developing branch of artificial intelligence known as machine learning aids with rainfall forecasting by **Vikas Kumar et al, (2022).** The UCI repository dataset with numerous characteristics will be used in this research study to forecast rainfall.

2.4 Yield prediction

Saeed Khaki and Lizhi Wang, (2021), One of the prevailing teams, created a deep neural network (DNN) strategy using state-of-the-art modelling and resolution techniques. In the validation dataset employing projected meteorological data, our model showed a greater degree of prediction accuracy, with a root-mean-square-error (RMSE) of 12% of the average

yield and 50% of the standard deviation. If the meteorological data were full, the RMSE would be reduced to 11% of the average yield and 46% of the standard deviation. The size of the input space was significantly decreased without notably affecting prediction accuracy by using the trained DNN model for feature selection. Our calculations demonstrate that this model significantly outperforms other well-known techniques like regression trees and shallow neural networks (SNN). After discovering them, we compiled the deep learning methods utilized in 30 of these papers. Convolutional Neural Networks (CNN) and Deep Neural Networks (DNN) are the two deep learning algorithms that are utilized in this study the most, with Long-Short Term Memory (LSTM) and Deep Neural Networks (DNN) being the other two. **Thomas van Klompenburga et al., (2020).**

Marizel B. Villanueva et al, (2018) This study looked at the bitter melon's or bitter gourd's ability to produce crops. The most popular term for it in the Filipino language is "Ampalaya." The primary source of data for the study was photographs of bitter melon leaves that were collected from farms in Ampalaya. The leaves were separated into groups for excellent and poor based on their description. Convolutional Neural Networks, a Machine Learning algorithm, were employed in the study. Data was trained using a combination of Python, TensorFlow, and Keras capabilities. In conclusion, by exposing a machine to a growing number of photos, it may eventually be able to distinguish between excellent and terrible Ampalaya plants.

Rushika Ghadge et al., (2018) employed supervised and unsupervised learning techniques, such as the Kohonen Self Organizing Map and the Back Propagation Network. In order to predict the kind of soil, learning networks are trained to classify the information into organic, inorganic, and real estate categories. Following a comparison of the accuracy produced using multiple network learning algorithms, the user is given the result that is the most correct. The system will evaluate the soil quality, forecast the crop output based on that evaluation, and provide fertilizer recommendations as necessary based on that evaluation.

Raghav et al., (2018) Based on the values gathered in real-time, the Machine Learning Algorithm (KNN) controls the parameter to recommend the crop that is optimal to grow in the specific field. For the purpose of predicting a crop, a standardized dataset that contains the essential elements for a certain crop is kept up to date. The field where the readings are required for calculation receives the sensors. The readings are transmitted in real-time to the cloud server using the DHT11, soil moisture sensor, MQ2, and light intensity sensor.

2.5 Summary

In the above, the literature says that smart farming and intelligent irrigation use IoT technology and artificial intelligence algorithm to improve productivity. IoT technology based on smart farming collects the sensor's data, such as temperature, humidity, moisture and pH, to update and monitor the data with the help of a dashboard. Machine learning and artificial intelligence involving in smart farming and irrigation system using various algorithms KNN, LSTM, OPNN and AJO, which are to analyse the data to predict the output as possible with precision, recall and accuracy. The data storage in the cloud is a big data storage system to analyze the data and update the dashboard with the help of an algorithm. So, we analyze the

above conversation to implement the intelligence irrigation and smart farming system, which will implement for visualising data NodeRED and Mosquito. And also, the above algorithm is to predict the output using a separate algorithm, but we have introduced an ensemble learning (SVM-KNN) algorithm to produce a result efficiently.

3. Methodology

3.1 AI-based smart irrigation system

Due to the utilization of intelligent machine-to-machine interaction in irrigation, precision agriculture has already made significant strides. The majority of IoT systems respond instantly and operate field valves at specific times. There is currently no system that makes decisions based on historical data and in-the-moment analysis. Only agricultural yielding, crop disease detection, and crop management issues have been solved using machine learning.

The ESP32 microcontroller will receive information about the root depth of the soil at regular intervals from temperature and moisture sensors. The 12-bit analog-to-digital converter included in the ESP32 microcontroller will transform the analog data input from the sensor to a digital one. Raspberry Pi will be linked to the esp32 microcontroller using the wireless sensor network concept. The computed temperature and soil moisture value will be sent to the raspberry Pi. The hardware employs an ensemble technique based on machine learning. This approach requires the coupling of two independent algorithms, such as SVM and KNN. Following applying the ensemble machine learning algorithm to the temperature and moisture readings, the Raspberry Pi3 instructs the ESP32 to turn on the irrigation pump connected to the relay based on the learned values. If the value meets the value necessary for the present soil demand, the pump won't get the command to begin the operation. No matter how much water is offered, the database is updated each time the algorithm examines the data. The number of irrigation operations and sensor readings are both included in this database. Farmers may access the data through a dashboard created using NodeRED using this database, which uploads itself to a website cloud gateway.

The fluctuation in conditions in prior years must be utilized to predict the future state of the rainfall. With characteristics including temperature, humidity, moisture sensor and SVM (support vector machine)-KNN (k-nearest neighbour algorithm) method has a very high probability that it will fit during the previous year's adjacent two-week period. As a result, this forecast will be accurate since it predicts rainfall based on historical data. The proposed intelligent irrigation system shows in Figure 1.

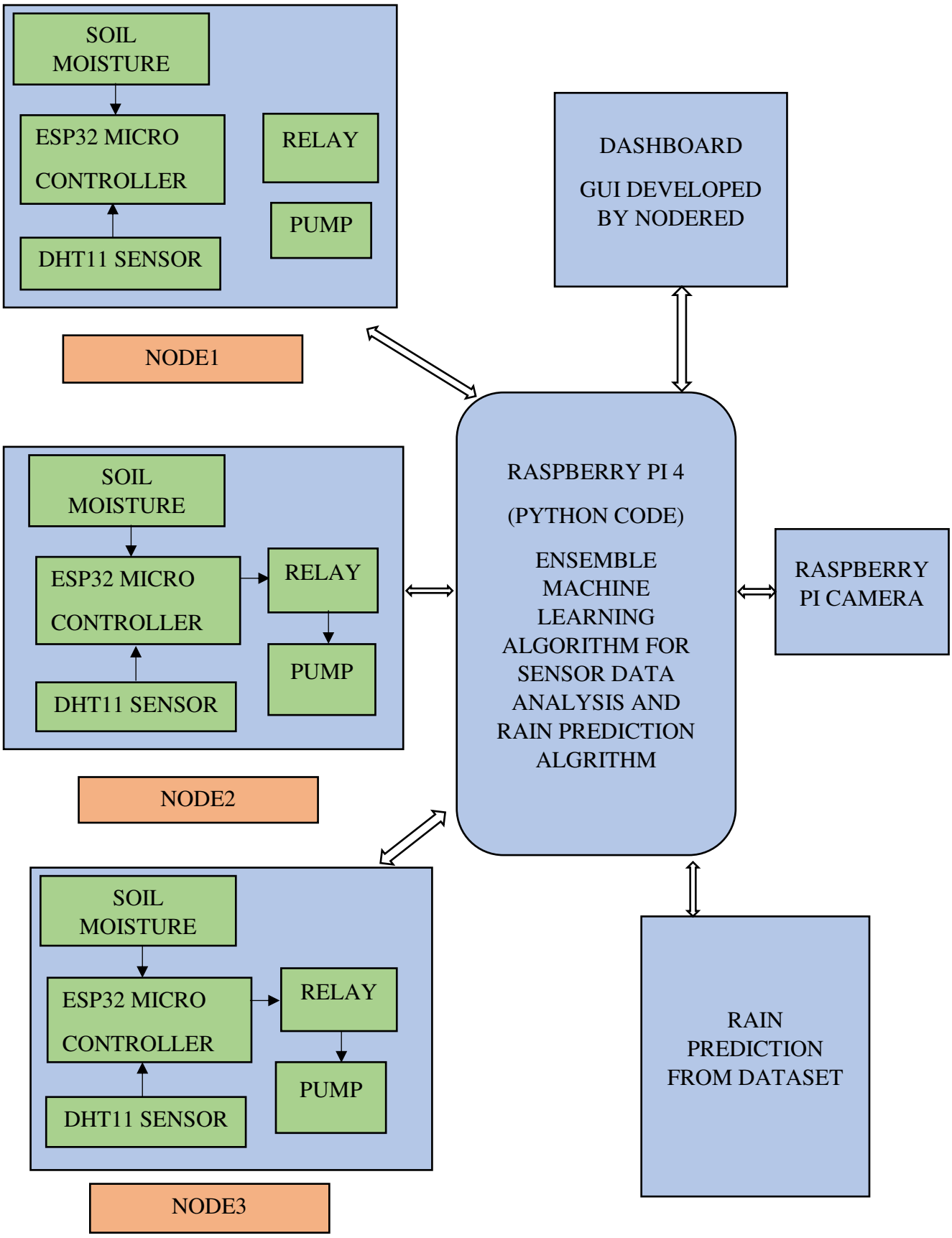


Figure 1. Proposed Intelligent Irrigation System Architecture

A centralised system controls the nodes and controls the operation of the pump attached to each node. A Raspberry Pi 4 running the Raspbian Operating System, which is based on Raspbian OS, serves as the solution's central controller. Using a Python3 script, the controller's functions are all carried out. When appropriate, the control logic uses methods imported by the main script from the support scripts. Interacting with the nodes, gathering data, and drawing conclusions using machine learning techniques are some of the features.

Each node is made up of a microcontroller board (ESP 32) based on the ESP8266 and many analogue sensors. The Analog Digital converters included into each ESP 32 translate the meteorological data collected by these sensors into digital 10-bit integers. An algorithm running on the ESP 32 transforms this input into the 32-bit floating point value needed by the inference model. Each node incorporates additional support hardware to provide the ESP 32 and the sensors with the required electrical power and to make straightforward troubleshooting possible in the event that the node malfunctions.

The duration is calculated based on the output of the regression method. The server gets a request from every node every 30 minutes. Thus, the network is essentially a wireless local area network. Since each node runs independently of the others, the system is not destroyed if a node accidentally fails. It is also easy to pinpoint the problematic node. Mosquitto is a broker for data transfer for web dashboard and it is a quick transfer of data in a dashboard.

3.2 Ensemble learning algorithm

Applying two or more models to the exact same data and aggregating the predictions from each model is a technique known as ensemble learning. Using the collection of models as a whole, ensemble learning is aimed at surpassing each individual model. To improve the performance of this task, intelligent irrigation and agricultural systems employ ensemble learning techniques like support vector machines (SVM) and the K-Nearest Neighbor algorithm (KNN).

3.2.1 Support vector machine (SVM)

Support vector machine is a popular machine learning algorithm and recent advances in statistical learning theory. The SVM follows the optimal hyperplane and satisfies the request for classification. After that, while guaranteeing the accuracy of the appropriate classification, employ a specific method to maximize the separation margin next to the suitable hyperplane. The several kinds of structural risk minimization depending on other learning algorithms have the ability to better generalization. The training dataset classifies the vector and label data which specifies the hyperplane sector. The training dataset consists of a number of data to be set $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, whereas, x_i-Input vector, x_i- Label. The mathematical expression of the partition hyperplane shows below,

$$\omega . x + b = 0 \quad (1)$$

Whereas,

b-offset hyperplane

ω- normal vector

The maximum partition hyperplane to reach a maximum at a bilateral blank area to choose the point at training dataset to make as possible, which defined as,

$$maximize\varphi(\omega) = \frac{1}{2} \parallel \omega \parallel^2 \quad (2)$$

To reach the constraint condition, which express,

$$y_i(\omega . x_i + b \geq 1) \quad (3)$$

The lagrange function is,

$$L(\omega, b, \alpha) = \frac{1}{2}(\omega . \omega) - \sum_{i=1}^{m} \alpha_i (y_i(\omega . x_i + b) - 1) \quad (4)$$

In above equations follow the two conditions, therefore, $\alpha_i \geq 0 \; and \; \sum_{i=1}^{m} y_i \alpha_i = 0$ to apply modify the equation in minimum of lagrange function as below,

$$maxQ(\alpha) = \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{m} \alpha_i \, \alpha_j y_i y_j (x_j . x_j) \quad (5)$$

The optimal class function as defined as,

$$f(x) = sgn\big((\omega^* . x) + b^*\big) = sgn(\sum_{i=1}^{n} \alpha_i^* \, y_i (x_i \, . x) + b^*) \quad (6)$$

The ability to mathematically examine SVM utilizing ideas from computational learning theory and achieve cutting-edge performance is a significant benefit. Recently, it has also been used to solve a variety of real-world issues, including the categorization of biological data and the recognition of handwritten characters. In order to improve the classification performance of agricultural data, SVM is presented in this research and used to categorize the data.

3.2.2 K-nearest neighbor algorithm

The k-nearest neighbor (k-NN) approach is a data mining methodology that is recognized as one of the top five data mining strategies, and it frequently uses the KNN Algorithm. Assuming that each of the qualities in our training program is a separate dimension in some space and that each observation's value for that feature is its coordinate in that dimension, the output is a set of points in space. Thus, the distance between two points in this space using any tenable metric may be used to determine how similar two points are to one another. The algorithm selects the k closest data points to determine which of the training set's points are similar enough to be taken into account when determining which class to forecast for a new observation. Because of its simplicity and speed of processing. In this work, the KNN classifier is calibrated using the Euclidean distance metric. The Euclidean distance calculation of the distance between the input vectors is given as,

$$D_{lj} = \sqrt{\sum_{k=1}^{n} \big(x_{ik} - x_{jk}\big)}, \; for \; k = 1,2,3, \ldots. \quad (7)$$

The Euclidean distance between a new point and the current point is calculated for each datum point in the dataset. The k elements with the most decreased separations to the new point will be selected after these separations are ordered in ascending order. The classifier identifies

the dominating class among these objects and then restores it as the class for the newly discovered point.

3.3 Rainfall prediction

The data set utilised for this study includes a wide range of factors in addition to the identified output class. A class that will be determined based on other available qualities is an output class. The output class was added to the dataset in order to assess the effectiveness and accuracy of machine-learning approaches. The outcome of the processing is comparable to the class that was identified, and performance is gauged by factors like accuracy, recall, and f1-score. The four stages of the classification model employed in this study are proper dataset selection, pre-processing, prediction, and outcomes simulation. The input data set for rainfall is drawn from the Tamilnadu Rainfall prediction website and includes a wide range of atmospheric characteristics. The accuracy of the outcome would suffer from a lack of information since a parameter with an incomplete value would be unable to participate in the prediction process. There was noise in the dataset, in addition to the missing values, when the value remained below or above certain boundaries. It is suggested that the values be kept within a certain range in order to get successful data mining outcomes.

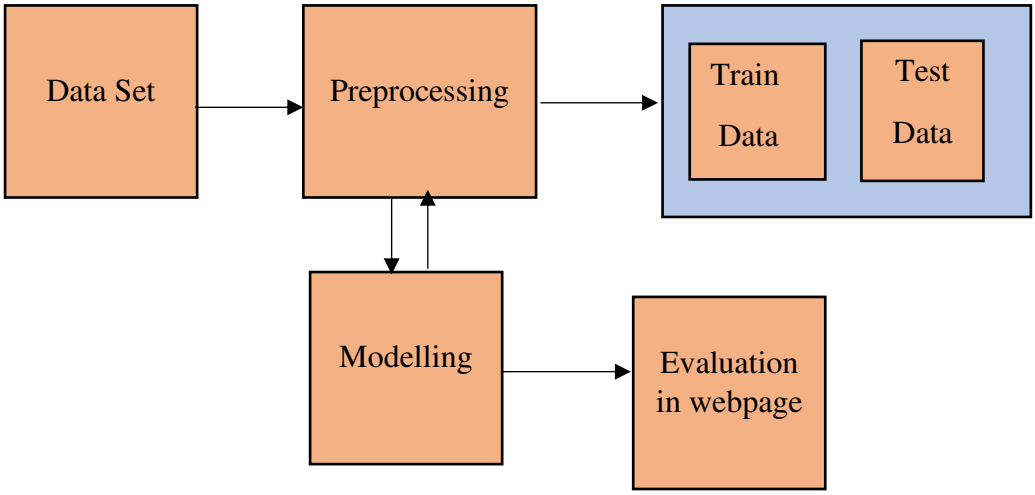


Figure 2. Rainfall prediction process

The dataset which contains the amount of data selected is probably not going to be in a format that enables working. Information cleaning also includes the completion of missing information or its removal. There may be instances where the data is lacking or partial, making it impossible to fix the issue. It could be necessary to eliminate these cases. More information may lead to computations taking longer to complete and using more memory and processing power. While analysing the complete dataset, it might do a smaller agent test on the chosen data, which might facilitate the study and testing of the setups. After preprocessing, steps involve the LSTM algorithm to evaluate the data from the dataset to choose the year and date and subsequently load the data on a webpage.

3.3.1 LSTM

LSTM is more accurate than traditional RNNs because of its long-range dependence. The RNN architecture's back-propagation algorithm causes the problem with error backflow. An LSTM's basic structure is referred to as a memory cell since it explicitly remembers and propagates unit outputs over a number of time steps. To store data regarding temporal contexts, the LSTM memory cell employs cell states. It also includes a forget gate, an input gate, and an output gate to control the information flow between various time steps. In this study, LSTM-based neural networks were trained to anticipate background radiation using time-series meteorological data. The mathematical challenge of comprehending the long-term interactions in the structure of recurrent neural networks is referred to as the "problem of vanishing gradient." As the input sequence gets longer, it gets harder to capture the impact of the previous phases. Gradients from the first few input points vanish and become zero. The activation function of the LSTM is thought of as the identity function with a derivative of 1.0 because of its recurrence nature. Consequently, the gradient that is being back-propagated doesn't disappear or explode; instead, it remains identical. The overall LSTM states and working of the sigmoid function, input layer, output layer and hidden layer depicts in Figure 3**.**

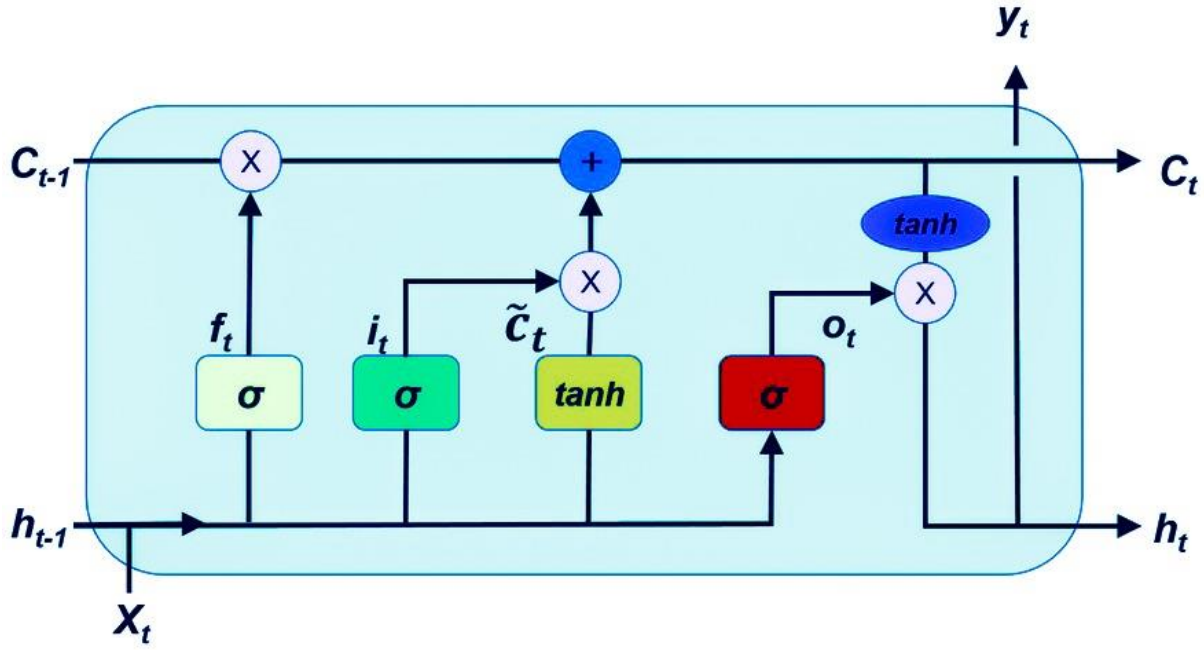


Figure 3. Architecture of LSTM

The hidden state expresses the mathematical formulae updated in the RNN algorithm shown below,

$$h_{(t)} = f(h_{t-1}, x_t) \quad (8)$$

Here, f- the activation function at non-linear

The memory cells maintain the linear unit with weight constant sector unit. The representation of the forget layer with the sigmoid function.

$$f_t = \sigma(W_{f.}[h_{t-1}, x_t] + b_f) \quad (9)$$

σ-sigmoid function,

x_t-input value,

W_f-weight of forgot gate,

b_f- bias value,

h_t- output value

The sigmoid function is,

$$\sigma(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (10)$$

The information that will be prevented is then decided by the model. The input gate layer and the sigmoid layer are the two layers that make up this procedure. Each layer serves to specify which values will be modified and creates a vector that contains potential new value possibilities. A new input value is produced when these two procedures are combined.

$$i_t = \sigma(W_i.[h_{t-1}, x_t] + b_i) \quad (11)$$

$$\bar{C}_t = tanh(W_c.[h_{t-1}, x_t] + b_c) \quad (12)$$

$$C_t = C_t * C_{t-1} + i_t * \bar{C}_t \quad (13)$$

Using a sigmoid gate to select the output value,

$$O_t = \sigma(W_O.[h_{t-1}, x_t] + b_O) \quad (14)$$

$$h_t = O_t * \tanh(C_t) \quad (15)$$

Whereas,

$tanh$-scale values range between -1to 1

W_i- weight input gate,

$\bar{C}_t$- added cell state,

C_t- cell state,

b_O- bias cell state,

O_t- output gate.

3.4 Yield prediction with machine learning and AI

The process of manually keeping track of agricultural activities and production patterns requires a lot of time and work. There is a good likelihood that farmers will only sometimes be around to monitor all agricultural activity. As a consequence, unpredictable and partial data are collected. AI-based solutions work well to handle such circumstances. This is essential in the technological age to encourage and prepare farmers to accept new tools and methods. This is extremely desirable to lessen physical labour, enhance crop quality, and fulfil the rising demand for food due to population growth. The problems encountered in conventional agriculture can be intelligently resolved through precision agriculture. The use of precision agriculture is being considered by farmers as a result of increased awareness and education. Farmers are open to implementing new precision agricultural techniques; however, accurate yield prediction is still difficult owing to form abnormalities, lighting circumstances, etc.

Precision agriculture involves tracking every crop's life cycle, from planting the seeds to harvesting them. It is an example of how a timetable for agricultural operations was planned using prior knowledge and data that had been gathered.

Crop images are first taken as part of the machine learning yield forecast process. Handheld cameras from mobile phones, digital cameras, satellite imaging, and other devices can capture images. Following capture, these images are utilized to train the appropriate neural networks. Predictions in a chosen domain are made using the trained networks. Retraining the model is necessary if the desired output and the ground truth don't match. Retraining continues until the model can forecast yields with accuracy.

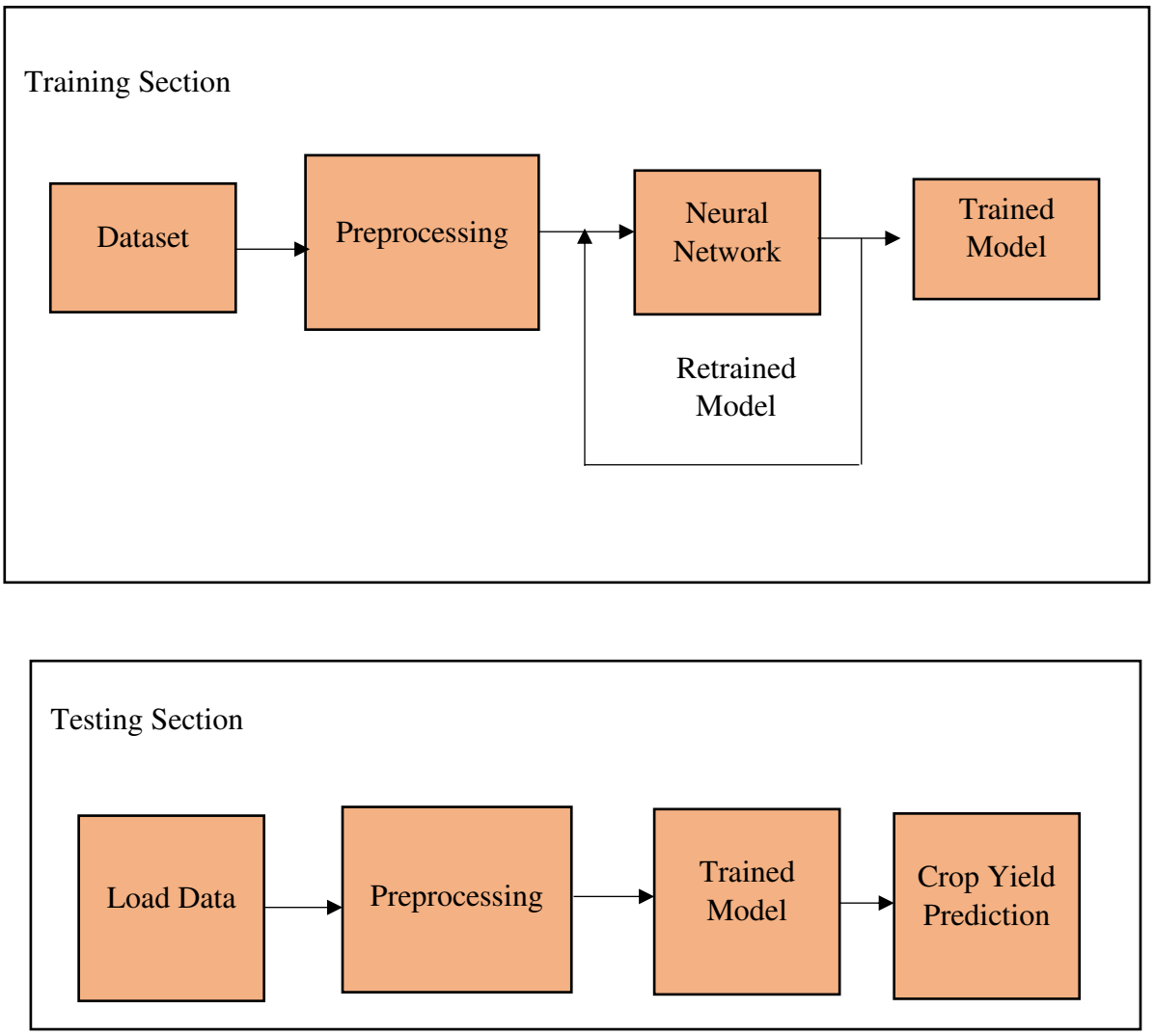
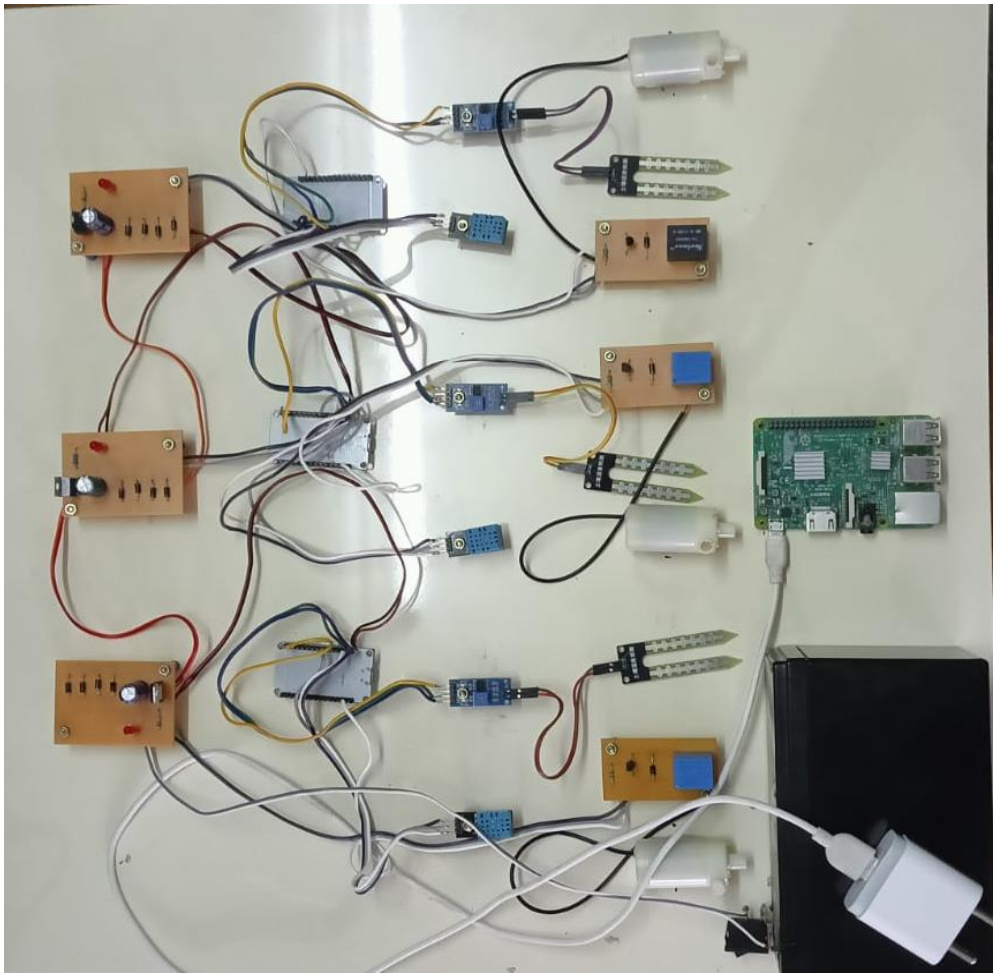


Figure 4. Flow diagram of yield prediction

4. Result and discussion

Smart farming and intelligent irrigation system using artificial intelligence systems and IoT developed with hardware and software contributions make an efficient performance. Our proposed method uses a soil moisture sensor and a DHT sensor to sense and evaluate values from soil. All hardware setup is shown in Figure 5.



This model is designed to find correlations between soil characteristics and crop yield. We may use this method to forecast how much of a certain crop will be produced. We are utilizing linear regression to determine the association between production and each of the factors. The linear equation is derived from a linear algorithm.

Coefficient of linear regression =β0x rainfall+β1x temperature +β2xpH

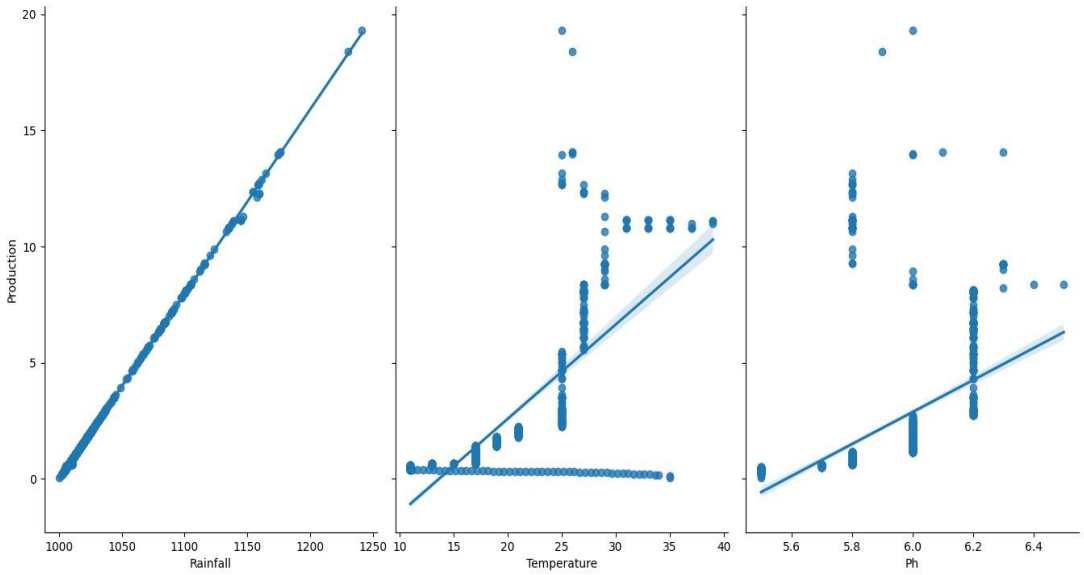


Figure 6. Analysis of rainfall, temperature and pH

Figure 6 represents the predicted values from the dataset, which is to produce the rainfall, temperature and pH using an ensemble learning algorithm in Python program language.

Together, all of these sensors help to provide real-time soil parameter information. We can irrigate the soil and forecast crops autonomously and their yield based on analyses performed using our machine learning algorithms, like linear regression.

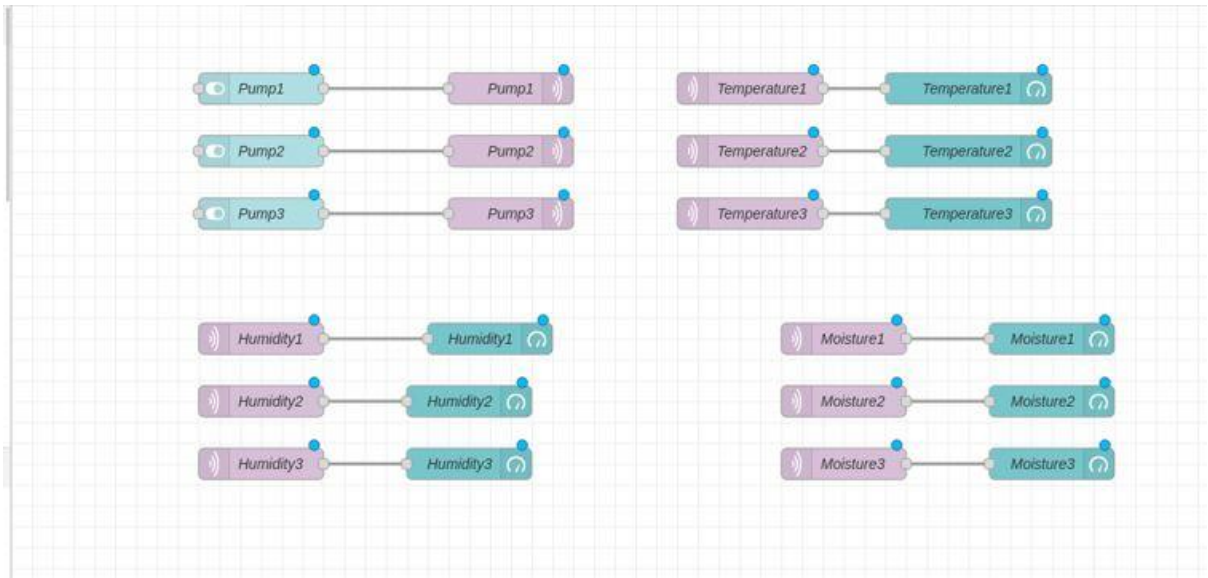


Figure 5. Performance of parameters and monitoring in a web dashboard

Figure 5 represents monitoring the temperature, humidity and moisture sensor values updated in the NodeRED webpage and transferring the messaging mosquito broker.

A farm has a network of 9 sensors spread across 3 parcels or cropped areas. The sensors collect information such as temperature and soil humidity. The objective is to foresee the parcels' water requirements. This is a classification issue with many labels. Each of the 20

sensors' observations in a multi-label classification task can have zero or more classifications assigned to it.

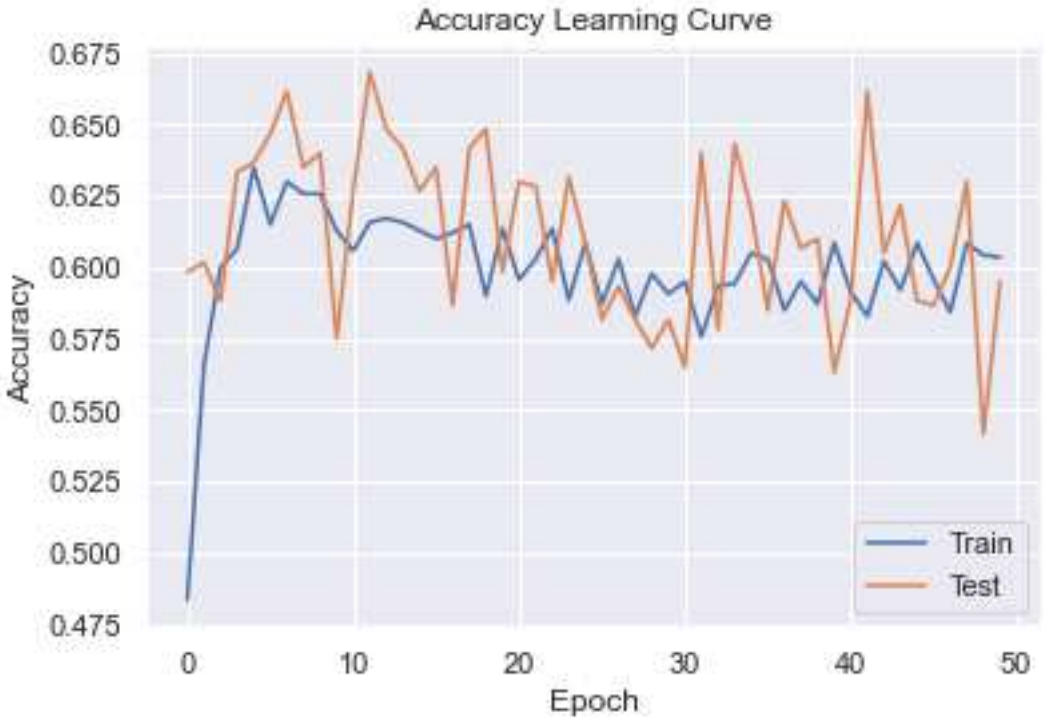


Figure 6. Accuracy learning curve

After the model changed the weights for the first few epochs, the validation accuracy is fairly excellent, between 0.85 and 0.90. Additionally, it appears that the validation curve's learning limit has been achieved using the supplied data whinch shows in Figure 6.

The train accuracy continues to rise after Epoch 18, while the validation accuracy stops rising at Epoch 18. This can be an overfitting sign. Let's add an EarlyStopping callback and increase the number of epochs to ensure this assumption and combat overfitting for a more reliable outcome.

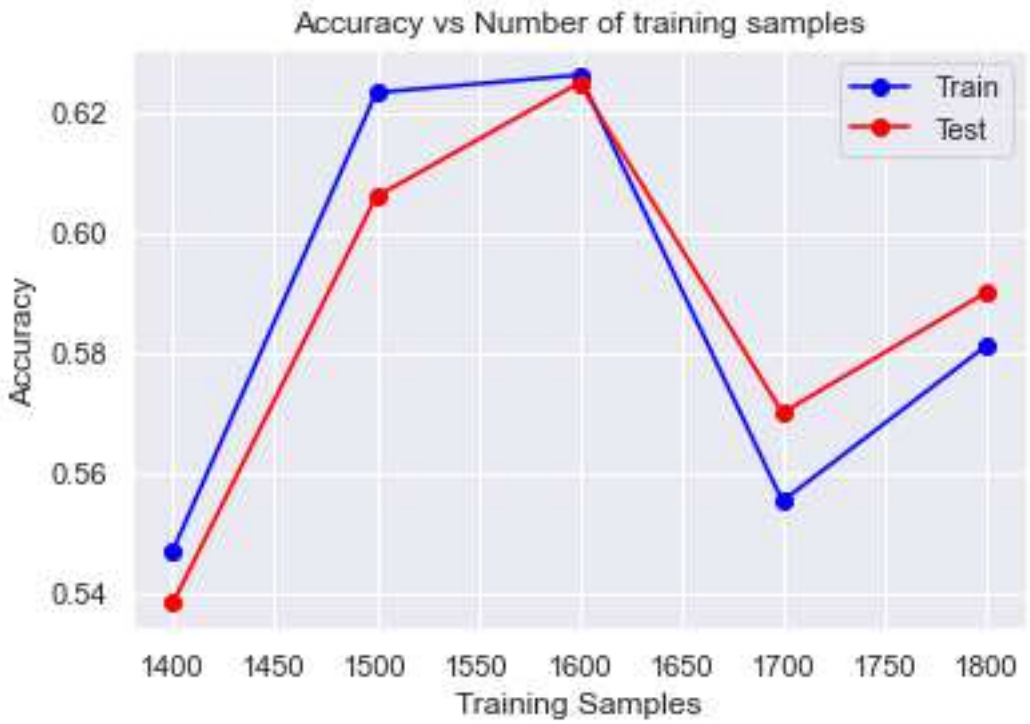


Figure 7. Accuracy obtained depends on number of training samples

Training the data with more than 1500 sensor observations reveals a negative trend, indicating that adding additional data won't improve the model's performance.

Table 1: Performance comparison of different algorithm

Algorithm	Accuracy
SVM	72%
KNN	76%
Ensemble learning algorithm (SVM-KNN)	90%

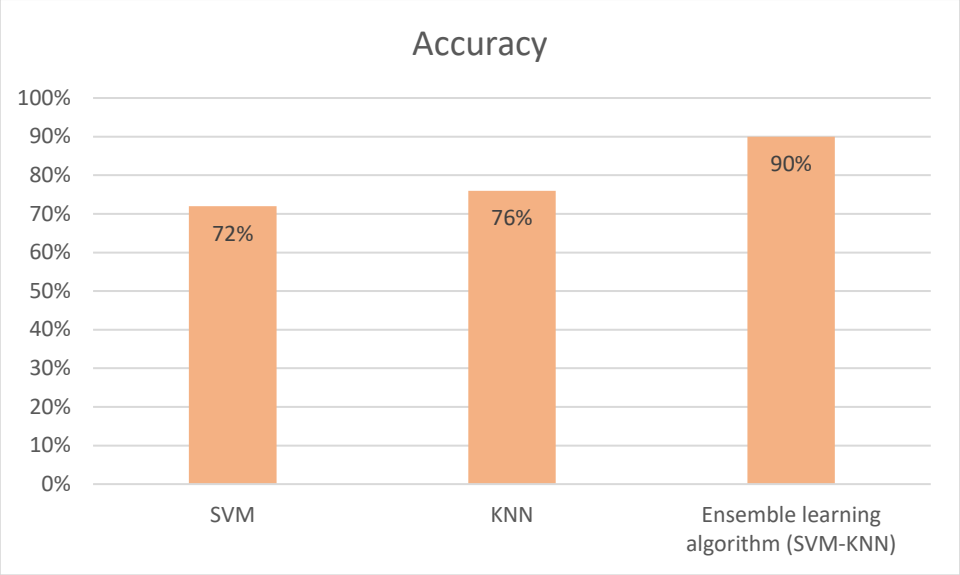


Figure 8. Comparison of algorithm

The ensemble learning algorithm decides the parameter which produces the accuracy was calculated percentage tabulated as in Table 1 and its comparison of plots shown in Figure 6, which is a better result in the ensemble learning algorithm.

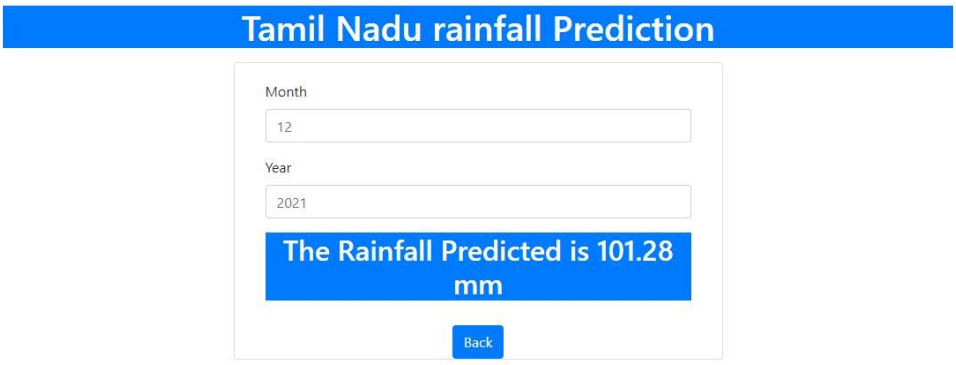


Figure 7. Webpage of rainfall prediction

5. Conclusion

Artificial intelligence-based smart farming and smart irrigation are discussed in this paper to provide better results. It also involves controlling how much water is used for irrigation. This article suggests an irrigation forecast that begins with developing a database utilizing a data acquisition card with several sensors and the Node-RED platform. This allowed us to gather various data for our machine learning-based decision support models. Precision agriculture is demanding in terms of intelligent irrigation systems and yield prediction. This chapter takes a thorough look at the important variables for estimating crop yields and irrigation needs. We have offered efficient prototypes and solutions after thoroughly studying several existing systems to fulfil the demands of the present environment. It has been discovered that AI-based algorithms are crucial in determining agricultural yields and irrigation requirements, working in concert with other hardware elements like the Raspberry Pi, soil moisture sensors, DHT-11 sensors, etc. If both yield prediction and intelligent irrigation systems are well-resourced through AI-based technology, they may demonstrate their efficacy in reducing total agricultural costs, promoting economic growth, and limiting the waste of necessary resources like water and energy. These methods help minimize human labor and accelerate the design of agricultural procedures. And also predict the rainfall from a database and provides a flow graph for specific conditions. The machine learning and artificial intelligence techniques used in smart farming and irrigation system for better accuracy value 72%, 76% and 90% using SVM, KNN and ensemble learning algorithm (SVM-KNN), respectively.

Declaration:

Ethics Approval and Consent to Participate:

No participation of humans takes place in this implementation process

Human and Animal Rights:

No violation of Human and Animal Rights is involved.

Funding:

No funding is involved in this work.

Data availability statement:

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study

Conflict of Interest:

Conflict of Interest is not applicable in this work.

Authorship contributions:

All authors are contributed equally to this work

Acknowledgement:

There is no acknowledgement involved in this work.

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